

Talking Points — Paper Tables, Run 20260817_015626 (corrected)

Companion to TABLES_20260817. For each table: the claim to make, the number to say out loud, what NOT to say, and the likely pushback with its answer. Prepared 2026-08-22.

The arc in one breath: (1) Tandem coupling is real and stress-ordered — T4, T16, T2/T3. (2) It has an identifiable mechanism — T6, T11b, T12. (3) Price discovery runs futures-first at machine timescales but the shares are near parity — T7, T13, T14, T8. (4) The measurement discipline is the differentiator — T15, T9, T10, T1.

T1 — Sample composition and 10 ms estimability

- **Say:** A designed sample, not a convenience sample: 25 matched pairs plus MWCB days, screened on realized volatility with the control picker blinded to outcomes. Data quality is audited per session, and we publish the roster (A1).
- **Say:** ES ladder staleness at 10 ms is early-era and loads on the *benchmark* side (64% vs 24% of regime). We measured the bias direction by planting known truth through the real pipeline: it attenuates. So the 10 ms regime gaps we report are conservative, not inflated.
- **Don't say:** "the 10 ms record is clean." The run estimated on sessions its own QC flags; the clean-subsample rerun and the dose-response covariate are the P0 checks on the next run.
- **Pushback:** "*Why not just drop the 25 stale sessions?*" Because they are 16/25 of the benchmark regime and almost all pre-2019 — exclusion destroys the era balance the pair design exists to protect, and power falls below 0.8. Direction-of-bias plus dose-response is the better instrument.

T2 — Table 5, crossed-book corner concentration

- **Say:** The corner concentration *orders* with stress: log OR $0.98 \rightarrow 1.12 \rightarrow 1.47$ from benchmark to volatile to MWCB. Positively co-moving flow corners are over-represented, negatively co-moving under-represented, and both margins steepen as markets degrade.
- **Say:** It is not a short-sale-restriction artifact — dropping the SSR member barely moves the MWCB panel ($1.470 \rightarrow 1.413$) — and the corner asymmetry is flat (≈ -0.02) in every panel.
- **Don't say:** any z or p from this table. The Woolf z's are pooled-bar statistics (T15 is the demonstration of why that disqualifies them). The claim, for now, is the monotone ordering plus T3's structure.
- **Pushback:** "*So Table 5 has no inference?*" Correct on this run, by our own hand: the day-clustered machinery (day bootstrap, within-pair sign-flip, MWCB jackknife) is built and attaches to the next run. We struck our own stars rather than let a referee do it.

T3 — Table 7, frequency-matched null

- **Say:** The published aggregation result reproduces on a fresh, designed sample: the negative-corner excess is 33.5x at 1 s bars, 1.32x at 10 ms, and 0.97x — nothing — in action time. Anti-correlated flow is a *clock artifact*; tandem (positively co-moving) flow is what survives at the transaction level.

- **Say:** This is the paper's cleanest replication-style exhibit and the best single argument for doing everything on two grids.
- **Don't say:** anything with a CI — none ships this run.

T4 — Tier-1 tandem flow coupling ★ the headline table

- **Say:** Flow coupling rises monotonically with stress on BOTH grids — benchmark $0.73 < \text{volatile } 0.82 < \text{MWCB } 0.87$ at 1 s; $0.27 < 0.29 < 0.35$ at 10 ms — and the volatile–benchmark difference is significant at both ($p = 0.0008$ and 0.0327 , day-level permutation).
- **Say:** This is the only flow exhibit that agrees across grids in both direction and significance; that cross-grid agreement is the reason it leads the results section.
- **Say (if pressed on staleness):** the measured bias is attenuation concentrated in the benchmark cells — the reported gap is a floor.
- **Don't say:** anything inferential about MWCB (4 days — describe, don't test).

T5 — Directional asymmetry

- **Say:** In calm regimes coupling tightens on down-moves; in volatile regimes it reverses to up-moves ($+0.047$, $p = 0.0045$ vs -0.090 , $p < 5e-5$ at 1 s). Candidate interpretation: hedging-driven coupling in calm markets, short-covering/rebound-chasing coupling in stressed ones.
- **Frame it as scale-dependent, on purpose:** the asymmetry lives at 1 s and vanishes at 10 ms — like Table 7, the phenomenon *emerges under aggregation*. That framing is consistent with the paper's thesis; hiding the 10 ms null would not be.
- **Don't say:** it is established. The gap-of-gaps is untested and the SSR/direction-conditioning checks are outstanding (P1).
- **Pushback:** "*Is the volatile reversal just SSR days?*" Unknown on this record — say so, and point to the planned conditioning check.

T6 — Mediation

- **Say:** The mediator link is the strongest coefficient in the run: a 1 SD innovation in tandem flow moves return correlation with $t \approx 18$, coefficient 67 (1 s) / 75 (10 ms). Whatever else is true, flow and correlation are one phenomenon at bar scale.
- **Say:** ES volatility suppresses flow innovations on both grids (Eq A negative twice). The full suppression decomposition — direct positive, indirect negative, total $\approx$ zero — holds at 1 s, where the indirect level CI $[-41.9, -3.1]$ excludes zero.
- **Don't say:** the share statistic (605% of a near-zero total — we suppressed it ourselves), and don't present the decomposition as grid-robust.
- **Pushback:** "*499 vs 199 bootstrap draws?*" Acknowledged imbalance; next run is 2000 draws with BCa intervals on both grids.

T7 — Lead-lag ★ the price-discovery table

- **Say (lead with the co-jump test):** When both assets jump, ES moves first on 65 of 66 days ($p = 5e-5$); the HRY timing estimate agrees — ES leads by ~ 20 ms, one grid step, on 64 of 66 days ($p = 0.0005$). Two independent day-level tests, same verdict.
- **Say:** It is NOT a staleness artifact — the lead is *weakest* on stale sessions (Spearman $+0.57$ with ES update counts), the opposite of what carry-forward contamination would produce.
- **Don't say:** θ magnitudes that include 2017-06-26 or 2017-10-05 (the two stalest days), and don't cite the pre-averaged θ column — it reverses sign and the contradiction is unadjudicated. The co-jump test carries the claim.
- **Say:** At 1 s the tests are silent ($p \approx 0.5, 0.96$) — the lead is a sub-100 ms phenomenon, which is exactly what T13 shows.

T8 — Information-share regime tests

- **Say:** The *level* result: information shares sit near parity — $ES \approx 0.46\text{--}0.47$ IS on average — against a literature prior of futures dominance. The ETF is a full partner in price discovery, not a satellite.
- **Say honestly:** the regime *shift* (ES gains share when volatile) is suggestive, not established: the one $p < 0.05$ cell (1 s free-permutation IS, $p = 0.032$) has no era-robust within-pair counterpart on this run — the within-pair test was run only on CS, where it dies. v0.9.87 adds the IS twin; the next run adjudicates.
- **Don't say:** "ES takes over in stress" — not on this record.
- **Pushback:** "Why trust IS over CS?" CS moves $+22\%$ across lag choices and swings 0.78 on one day under beta re-estimation; IS moves -0.014 and 0.127 respectively. Fragility is measured, not asserted.

T9 — Table 9, state-dependent IRFs

- **Say:** Measurement choice changes conclusions: under HY (Epps-free), the ES weighted-spread response flips sign vs Pearson (-0.048^{**} vs $+0.001$), i.e. part of the published response is measurement artifact — the Δ column quantifies exactly how much, shock by shock.
- **Say:** RV shocks dominate the correlation response in both regimes.
- **Don't interpret:** DCC's near-zero responses ($a+b = 0.9999$, boundary estimate) or anything at 10 ms (that leg died at ~ 293 GiB and will return memory-bounded), and no cell is lag-band-certified this run.
- **Position:** this table is transitional — the lag-order-free local projection twin (T9_LP=1) supersedes it next run.

T10 — FEVD/GFEVD

- **Say:** Roughly half of SPY-return forecast-error variance attributes to ES order flow (GFEVD: 46% benchmark, 50% volatile). For an instrument with its own \$500B book, that is the tandem-market thesis in one number.
- **Say:** The diagonal-vs-generalized gap is itself evidence: orthogonal- shock FEVD fails precisely because flow innovations correlate at $0.70\text{--}0.79$ — you cannot even *decompose* variance while pretending the legs

are independent.

- **Caveat to volunteer:** under tandem flow, common flow is counted toward both shocks — the GFEVD shares are upper bounds on unique attribution.
- **Note:** regime labels here were restored from the run report; the shipped CSVs were unlabeled (disclosed in the table note).

T11a — Copula tail dependence by regime

- **Say:** Tail dependence rises from calm to stressed days (median $\lambda \approx 0.53 \rightarrow 0.69$), and the winning copula family *changes* — symmetric t on calm days, asymmetric SJC/BB1 on volatile days. Stress does not just raise co-movement; it reshapes the joint tail.
- **Don't say:** benchmark tails are symmetric — the t family *forces* $\lambda_L = \lambda_U$ and t wins 19/25 benchmark days, so that symmetry is partly assumption. The nonparametric estimator (v0.9.87) adjudicates next run.
- **Note:** the upper-tail LR column is omitted pending its definition fix.

T11b — Tail dependence, thin vs deep books ★ the mechanism table

- **Say:** Within volatile days, lower-tail dependence is significantly higher when the book is thin (+0.076, $p = 5e-5$). Liquidity state is not a backdrop — it is the transmission channel: thin books are when crashes travel together.
- **Say:** This is arguably the cleanest mechanism result in the paper — within-day, within-regime, day-level inference.

T12 — Markov-switching flow regimes

- **Say:** Coupling is not a daily constant: 64 of 66 days reject a single within-day regime. Calm days are long low-coupling spells punctuated by short high-coupling bursts (median 156 vs 36 bars, asymmetry $p = 0.0007$); volatile days live in the coupled state.
- **Say:** Regime entries lean toward down-moves ($p = 0.020$ pooled) — coupling switches on when prices fall.
- **Phrase carefully:** "rejected at the bootstrap's resolution floor" (99 draws $\rightarrow p = 0.01$ is a floor, not an estimate).

T13 — Horizon profile

- **Say:** The cross-impact asymmetry survives every sampling rung — $\lambda_{SPY \leftarrow ES}$ is an order of magnitude above $\lambda_{ES \leftarrow SPY}$ from 10 ms to 1 s — while the ES lead share is visible only below 100 ms. The lead is fast; the asymmetry is structural.
- **Say (proactively):** we caught and fixed a units error here — the shipped half-life column was in bars; corrected, error-correction half-life runs 0.05 s at 10 ms to ~64 s at 1 s. Volunteering this buys credibility for the rest.

- **Don't say:** level comparisons of λ across rungs as if they were economics — bar aggregation moves them mechanically; the *ordering within rung* is the content.

T14 — ECM-SDE

- **Say:** Textbook error correction with the right signs on both grids: SPY adjusts toward the common curve, ES pulls it — consistent with T7's lead. Microprice-based IS exceeds mid-based (0.65 vs 0.57 at 10 ms): book imbalance carries information beyond the mid.
- **Say:** At 1 s the SPY adjustment coefficient is already insignificant ($t = 0.30$) — by one second, the correction is done. Same story as T13.

T15 — iid vs day-clustered ★ the referee-defense table

- **Say:** Same regression, same data: pooled-iid $t = 386.9$, day-clustered $t = 5.56$ — a factor of 69.5. This is the paper's inference policy in one row: nothing pooled gets stars, everything is day-level or day-clustered.
- **Use it** to preempt the " $n = 1.5$ million" objection: the effective n is 66, and we act like it. It is also the stated reason T2's z column is struck — the discipline is applied to our own headline table first.

T16 — Cross-impact summary

- **Say:** Cross-impact triples from benchmark to volatile ($0.012 \rightarrow 0.035$) while own-impact barely moves ($0.84 \rightarrow 0.79$). The tandem channel is what strengthens under stress — not markets generally becoming more impactful.
- **Note:** "volatile" here includes the MWCB days (folded taxonomy, marked).

A1 — Per-session QC roster

- **Say:** Every session's staleness fraction and QC verdict is published; any reader can recompute any table excluding any session. The two known defects (2017-06-26 ladder monotonicity; 2020-12-11 contract-identity scoring) are disclosed in place.
 - **Use it** as the transparency close: the paper's data-quality argument is auditable, not rhetorical.
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If you get one skeptical question per section

- *Results:* "Isn't the 10 ms benchmark cell garbage?" → T1: bias direction measured (attenuation), gaps conservative; dose-response next run.
- *Inference:* "Where are Table 5's stars?" → T15: we struck them ourselves; day-clustered replacements are built and attach to the next run.
- *Price discovery:* "20 ms lead with stale data?" → T7: the lead is weakest where data is stalest; artifact would run the other way.

- *Novelty*: "What's actually new?" → T4 + T11b + T3: stress-ordered coupling measured at flow level on a validated two-grid tape, with a liquidity-state mechanism and a clock-artifact decomposition nobody else has run.